

5

4

3

2

1

W101-001 REV A

NOTES:

1. INTERPRET DRAWING PER ASME Y14.100.

2. DIMENSIONS ARE IN INCHES. LENGTH TOLERANCE ±0.5 IN.

3. WIRE PER WIRE LIST. TERMINATE CONTACTS PER CONNECTOR MANUFACTURER'S INSTRUCTIONS.

4. 100% CONTINUITY AND ISOLATION TEST PER WIRE LIST.

5. ASSEMBLY VIEW AND CONNECTOR PINOUTS ON SHEET 2. WIRING DIAGRAM ON SHEET 3. WIRE LIST, WIRE GROUPS, SPLICES AND LABEL SCHEDULE ON SHEET 4.

6. DIMENSIONS PER IPC-D-620: HARNESS DIMENSIONS, INCLUDING BREAKOUT AND SPLICE LOCATIONS, ARE MEASURED FROM DATUM A, THE P1 CONNECTOR FACE. HARNESS LENGTH IS MEASURED FROM DATUM A TO THE FINAL TERMINATION. BREAKOUT LENGTHS ARE MEASURED FROM THE APPROXIMATE CENTERLINE OF THE HARNESS AT THE BREAKOUT TO THE FINAL TERMINATION.

7. CONTACTS (M39029/56-351, M39029/58-363) ARE SUPPLIED WITH CONNECTORS (FIND NOS. 1, 2); DO NOT ORDER SEPARATELY.

8. CONNECTOR PINOUTS SHOW THE FRONT (ENGAGING) FACE OF THE CONNECTOR CALLED OUT: INSERT ARRANGEMENT PER MIL-STD-1560, SOCKET INSERTS AS THE MIRROR IMAGE OF THE PIN INSERT. FILLED CAVITIES ARE WIRED; OPEN CAVITIES ARE NC. MASTER KEYWAY NOT SHOWN.

9. NC (NO CONNECTION) CONTACT POSITIONS: P1: N, P, R, S, T, U; P2: N, P, R, S, T, U.

10. TWIST THE WIRES OF EACH TWISTED GROUP TOGETHER OVER THEIR FULL LENGTH (SEE WIRE GROUPS TABLE).

11. CALCULATED BUNDLE DIAMETER AT CONNECTORS (REF): P1 0.257 IN, P2 0.257 IN.

12. TERMINATE SHIELDS PER WIRE GROUPS TABLE USING FIND NO. 13. INSULATE FLOATING SHIELD ENDS.

13. INSTALL SPLICE (FIND NO. 14) AT LOCATION SHOWN; SEE SPLICE TABLE.

14. RECOVER HEATSHRINK BOOT (FIND NO. 4) OVER BACKSHELL CABLE CLAMP.

15. INSTALL IDENTIFICATION LABEL (FIND NO. 5) ADJACENT TO CONNECTOR; MARK PER LABEL SCHEDULE.

16. INSTALL WIRE MARKER (FIND NO. 15) AT EACH END OF EACH WIRE; MARK WITH WIRE ID.

REVISIONS

ZONE

REV

DESCRIPTION

DATE

APPROVED

A

INITIAL RELEASE

2026-10-08

J. SMITH

15

26

TMS-SCE-1/8-2.0-9

MARKER, WIRE, 1/8 IN

16

14

1

M81824/1-1

SPLICE, CRIMP, ENVIRONMENT RESISTANT

13

13

2

M83519/2-3

SHIELD TERMINATION, SOLDER SLEEVE

12

12

48 IN

M22759/16-20-5

WIRE, 20 AWG, GRN

11

54 IN

M22759/16-22-0

WIRE, 22 AWG, BLK

10

48 IN

M22759/16-22-6

WIRE, 22 AWG, BLU

9

AR

M22759/16-22-9

WIRE, 22 AWG, WHT

8

48 IN

M22759/16-20-0

WIRE, 20 AWG, BLK

7

48 IN

M22759/16-20-2

WIRE, 20 AWG, RED

6

48 IN

M27500-22TG2T14

CABLE, SHIELDED TWISTED PAIR, 22 AWG

5

2

TMS-SCE-1/2-2.0-9

SLEEVE, IDENTIFICATION, 1/2 IN

15

4

2

202K153-25/225-0

BOOT, HEATSHRINK, STRAIGHT

14

3

2

M85049/38S15W

BACKSHELL, STRAIGHT, SHELL 15

2

1

D38999/26WD18PN

CONNECTOR, PLUG, 18 PIN

1

1

D38999/26WD18SN

CONNECTOR, PLUG, 18 SKT

FIND NO

QTY REQD

CAGE CODE

PART OR IDENTIFYING NO

NOMENCLATURE OR DESCRIPTION

NOTE

PARTS LIST

APPLICATION

NEXT ASSY

USED ON

A1000-001

FCC INST...

UNLESS OTHERWISE SPECIFIED:

DIMENSIONS ARE IN INCHES

LENGTH TOLERANCE ±0.5

INTERPRET PER ASME Y14.5

DO NOT SCALE DRAWING

THIRD ANGLE PROJECTION

CONTRACT NO.

APPROVALS

DATE

DRAWN

K. KIPER

CHECKED

J. SMITH

ENGR

2026-10-08

2026-10-08

APPROVED

ACME AEROSPACE (EXAMPLE)

TITLE

CABLE ASSY, FCC TO PDU

SIZE

CAGE CODE

DWG NO.

REV

C

W101-001

A

SCALE

WEIGHT

SHEET

NONE

1 OF 4

5

4

3

2

1

5

4

3

2

1

W101-001 REV A

ASSEMBLY VIEW AND CONNECTOR PINOUTS (NOT TO SCALE)

C

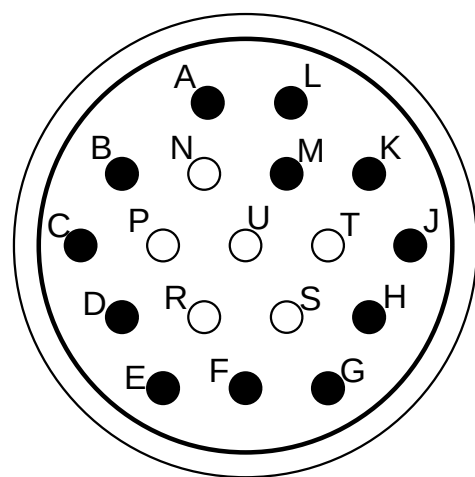
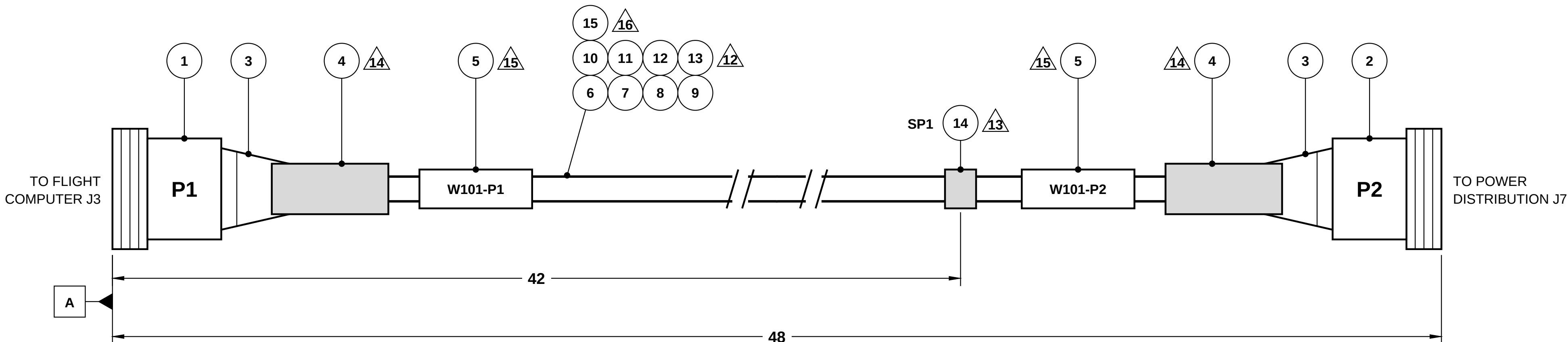
C

B

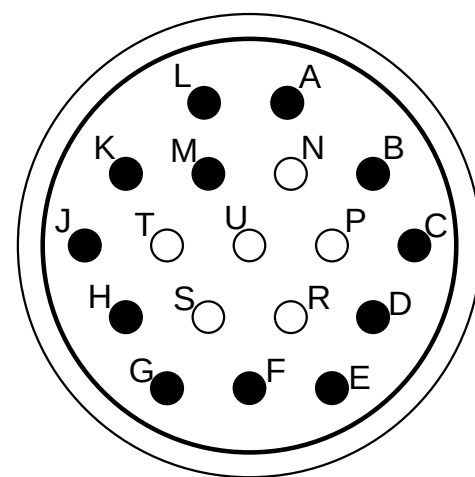
B

A

A



P1 PINOUT, FRONT FACE
D38999/26WD18SN
INSERT 15-18, KEY N
SOCKET CONTACTS



P2 PINOUT, FRONT FACE
D38999/26WD18PN
INSERT 15-18, KEY N
PIN CONTACTS

SIZE C	CAGE CODE	DWG NO. W101-001	REV A
SCALE NONE	WEIGHT	SHEET 2 OF 4	

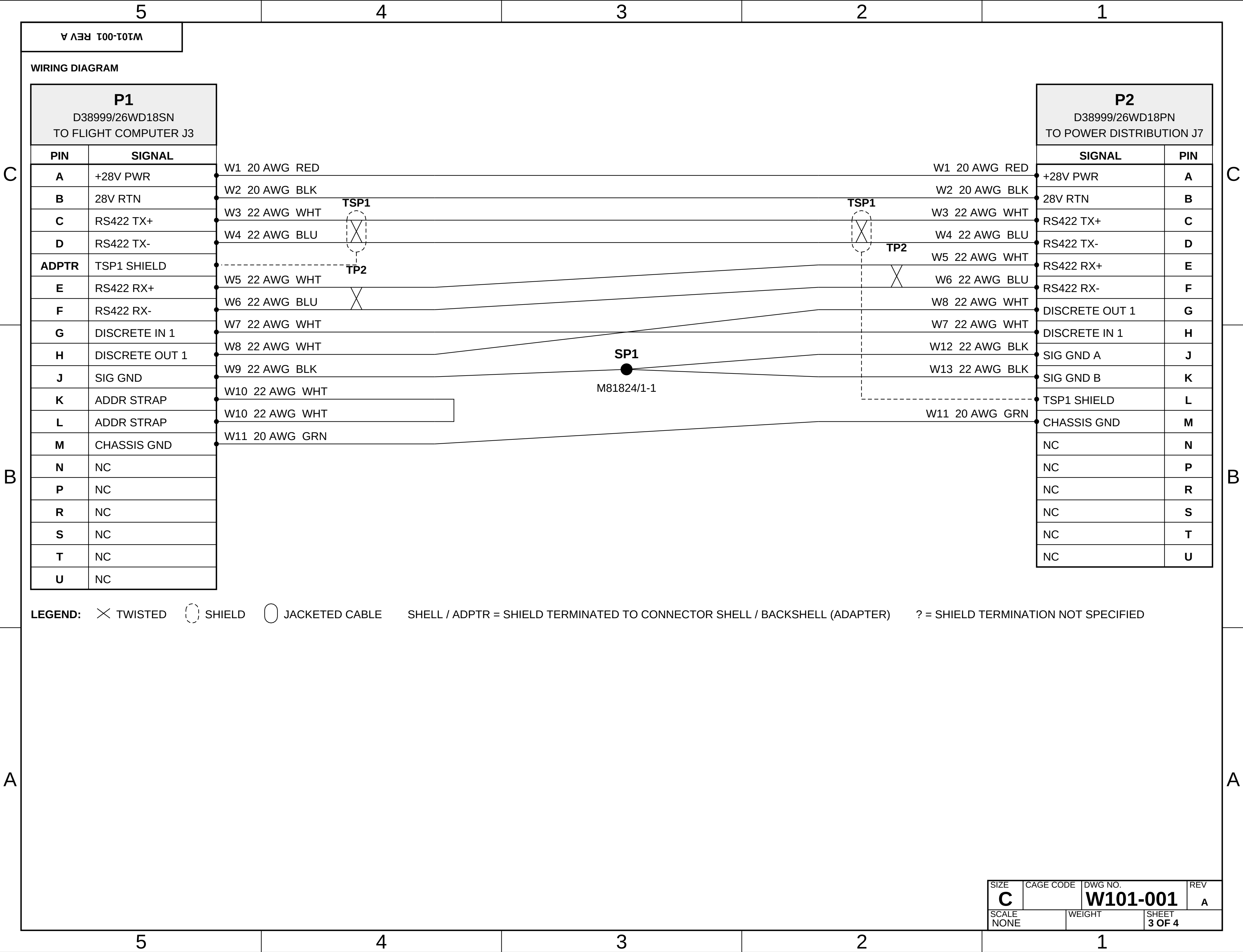
5

4

3

2

1



5

4

3

2

1

W101-001 REV A

WIRE LIST AND TABLES

WIRE LIST

WIRE	FROM	PIN	TO	PIN	SIGNAL	AWG	COLOR	WIRE P/N	GROUP	LENGTH (IN)	NOTES
W1	P1	A	P2	A	+28V PWR	20	RED	M22759/16-20-2		48	
W2	P1	B	P2	B	28V RTN	20	BLK	M22759/16-20-0		48	
W3	P1	C	P2	C	RS422 TX+	22	WHT	(M27500-22TG2T14)	TSP1	48	
W4	P1	D	P2	D	RS422 TX-	22	BLU	(M27500-22TG2T14)	TSP1	48	
W5	P1	E	P2	E	RS422 RX+	22	WHT	M22759/16-22-9	TP2	48	
W6	P1	F	P2	F	RS422 RX-	22	BLU	M22759/16-22-6	TP2	48	
W7	P1	G	P2	H	DISCRETE IN 1	22	WHT	M22759/16-22-9		48	
W8	P1	H	P2	G	DISCRETE OUT 1	22	WHT	M22759/16-22-9		48	
W9	P1	J	SP1		SIG GND	22	BLK	M22759/16-22-0		42	
W10	P1	K	P1	L	ADDR STRAP	22	WHT	M22759/16-22-9		AR	JUMPER IN P1
W11	P1	M	P2	M	CHASSIS GND	20	GRN	M22759/16-20-5		48	
W12	SP1		P2	J	SIG GND A	22	BLK	M22759/16-22-0		6	
W13	SP1		P2	K	SIG GND B	22	BLK	M22759/16-22-0		6	

WIRE GROUPS AND SHIELDS

GROUP	TYPE	WITHIN	WIRES	CABLE P/N	SHIELD P/N	SHIELD TERM	SHIELD (FROM END)	SHIELD (TO END)	NOTES
TSP1	SHIELDED TWISTED PAIR		W3, W4	M27500-22TG2T14		M83519/2-3 (FIND NO. 13)	P1: BACKSHELL	P2: PIN L	RS422 TX
TP2	TWISTED PAIR		W5, W6						RS422 RX; 10 TWISTS PER FOO...

SPLICES

SPLICE	FIND NO	SPLICE P/N	LOCATION	WIRES	NOTES
SP1	14	M81824/1-1	42 IN FROM DATUM A (P1 FACE)	W9, W12, W13	SIGNAL GROUND SPLICE

LABEL SCHEDULE

FIND NO	LABEL P/N	LEGEND	LOCATION	QTY
5	TMS-SCE-1/2-2.0-9	W101-P1	CABLE, ADJACENT TO P1	1
5	TMS-SCE-1/2-2.0-9	W101-P2	CABLE, ADJACENT TO P2	1
15	TMS-SCE-1/8-2.0-9	W1	WIRE W1, AT P1-A AND P2-A	2
15	TMS-SCE-1/8-2.0-9	W2	WIRE W2, AT P1-B AND P2-B	2
15	TMS-SCE-1/8-2.0-9	W3	WIRE W3, AT P1-C AND P2-C	2
15	TMS-SCE-1/8-2.0-9	W4	WIRE W4, AT P1-D AND P2-D	2
15	TMS-SCE-1/8-2.0-9	W5	WIRE W5, AT P1-E AND P2-E	2
15	TMS-SCE-1/8-2.0-9	W6	WIRE W6, AT P1-F AND P2-F	2
15	TMS-SCE-1/8-2.0-9	W7	WIRE W7, AT P1-G AND P2-H	2
15	TMS-SCE-1/8-2.0-9	W8	WIRE W8, AT P1-H AND P2-G	2
15	TMS-SCE-1/8-2.0-9	W9	WIRE W9, AT P1-J AND SP1	2
15	TMS-SCE-1/8-2.0-9	W10	WIRE W10, AT P1-K AND P1-L	2
15	TMS-SCE-1/8-2.0-9	W11	WIRE W11, AT P1-M AND P2-M	2
15	TMS-SCE-1/8-2.0-9	W12	WIRE W12, AT SP1 AND P2-J	2
15	TMS-SCE-1/8-2.0-9	W13	WIRE W13, AT SP1 AND P2-K	2

SIZE
C

SCALE
NONE

CAGE CODE

WEIGHT

DWG NO.
W101-001

SHEET
4 OF 4

REV
A

5

4

3

2

1